EXPERIMENT – 4

AIM: To perform and analyze SQL queries using INNER JOIN, LEFT JOIN, SELF JOIN, 3-way JOIN, correlated subqueries, EXISTS, and simulated INTERSECT and EXCEPT operations on the Employee–Department–Project database, and to compare their execution plans using EXPLAIN.

SOURCE CODE:

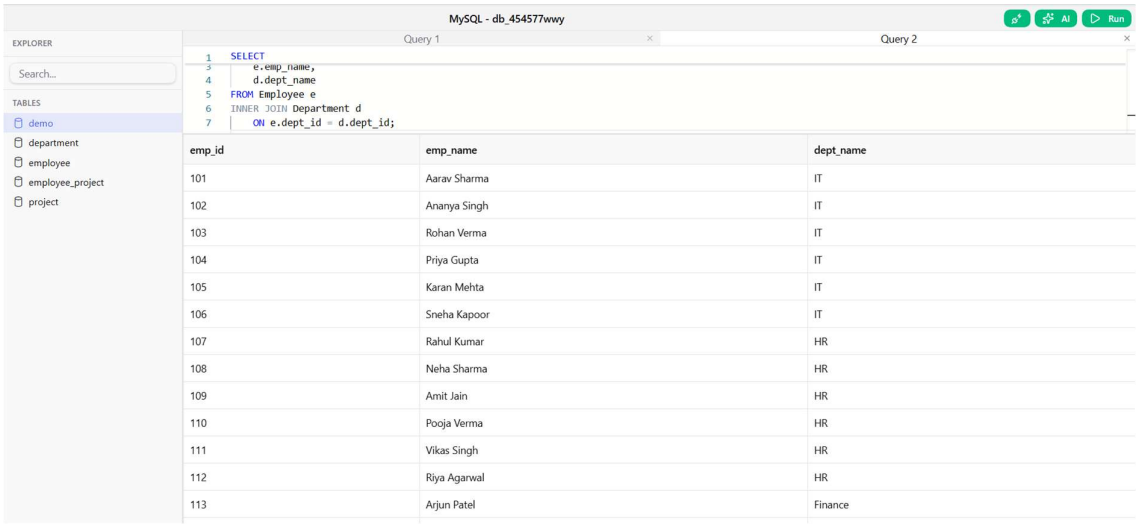
USING EXPERIMENT – 3 DATABASE

SQL QUERIES:

INNER JOIN

```
SELECT
    e.emp_id,
    e.emp_name,
    d.dept_name
FROM Employee e
INNER JOIN Department d
    ON e.dept_id = d.dept_id;
```

OUTPUT:

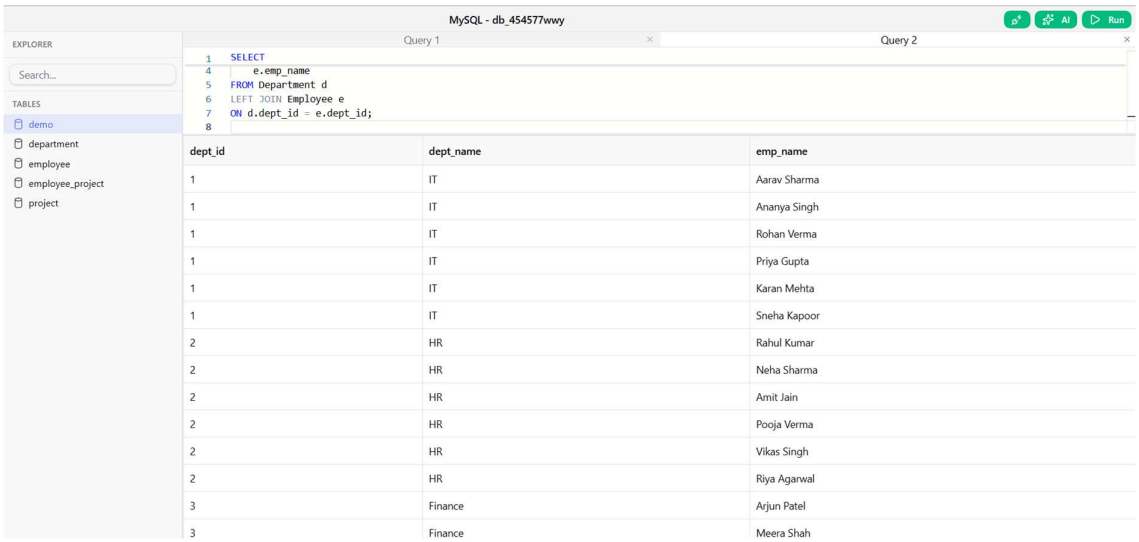


emp_id	emp_name	dept_name
101	Aarav Sharma	IT
102	Ananya Singh	IT
103	Rohan Verma	IT
104	Priya Gupta	IT
105	Karan Mehta	IT
106	Sneha Kapoor	IT
107	Rahul Kumar	HR
108	Neha Sharma	HR
109	Amit Jain	HR
110	Pooja Verma	HR
111	Vikas Singh	HR
112	Riya Agarwal	HR
113	Arjun Patel	Finance

LEFT JOIN

```
SELECT
    d.dept_id,
    d.dept_name,
    e.emp_name
FROM Department d
LEFT JOIN Employee e
    ON d.dept_id = e.dept_id;
```

OUTPUT:

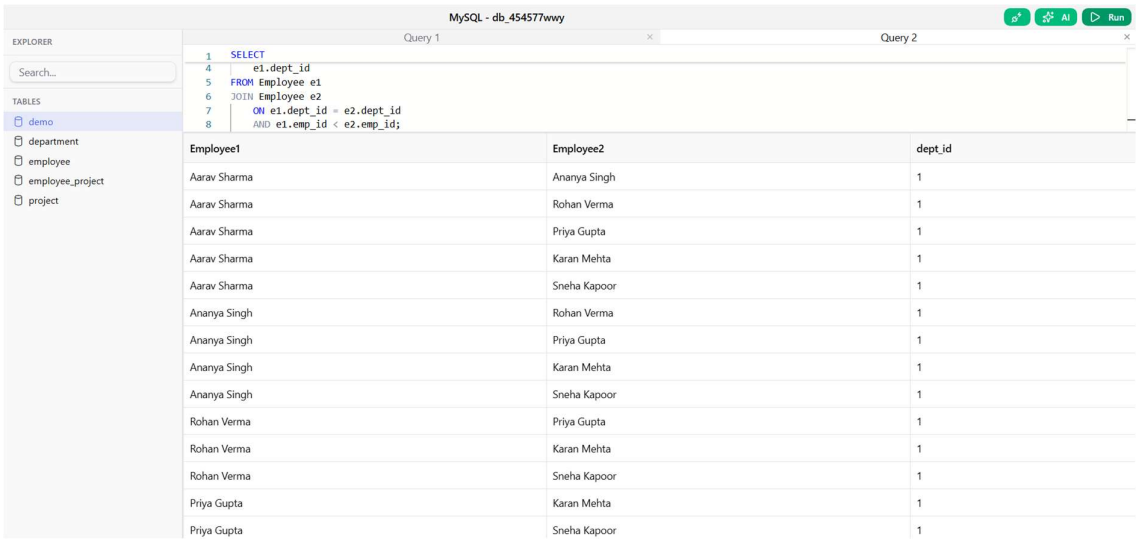


dept_id	dept_name	emp_name
1	IT	Aarav Sharma
1	IT	Ananya Singh
1	IT	Rohan Verma
1	IT	Priya Gupta
1	IT	Karan Mehta
1	IT	Sneha Kapoor
2	HR	Rahul Kumar
2	HR	Neha Sharma
2	HR	Amit Jain
2	HR	Pooja Verma
2	HR	Vikas Singh
2	HR	Riya Agarwal
3	Finance	Arjun Patel
3	Finance	Meera Shah

SELF JOIN

```
SELECT
    e1.emp_name AS Employee1,
    e2.emp_name AS Employee2,
    e1.dept_id
FROM Employee e1
JOIN Employee e2
    ON e1.dept_id = e2.dept_id
    AND e1.emp_id < e2.emp_id;
```

OUTPUT:

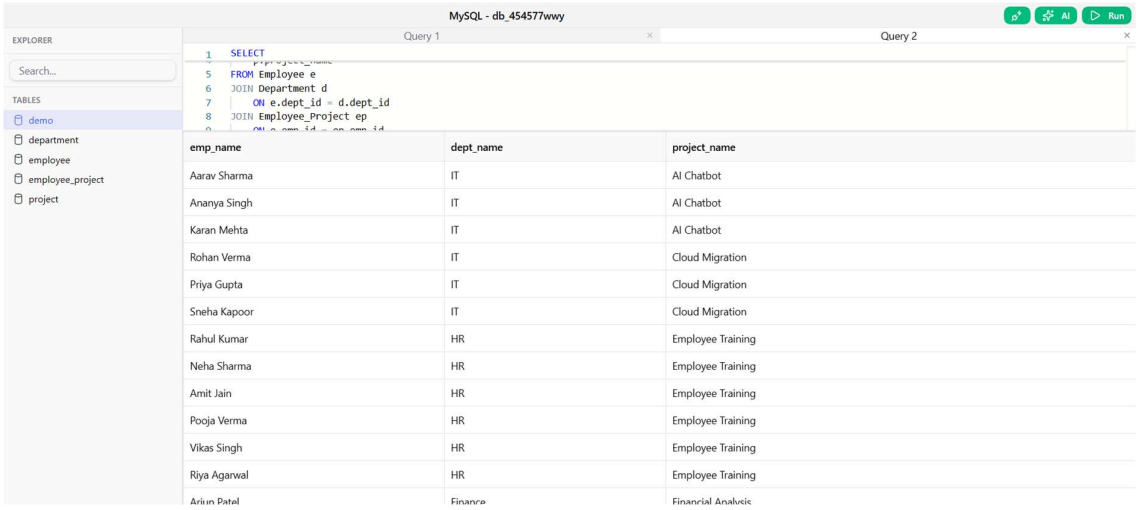


Employee1	Employee2	dept_id
Aarav Sharma	Ananya Singh	1
Aarav Sharma	Rohan Verma	1
Aarav Sharma	Priya Gupta	1
Aarav Sharma	Karan Mehta	1
Aarav Sharma	Sneha Kapoor	1
Ananya Singh	Rohan Verma	1
Ananya Singh	Priya Gupta	1
Ananya Singh	Karan Mehta	1
Ananya Singh	Sneha Kapoor	1
Rohan Verma	Priya Gupta	1
Rohan Verma	Karan Mehta	1
Rohan Verma	Sneha Kapoor	1
Priya Gupta	Karan Mehta	1
Priya Gupta	Sneha Kapoor	1

3-WAY JOIN

```
SELECT
    e.emp_name,
    d.dept_name,
    p.project_name
FROM Employee e
JOIN Department d
    ON e.dept_id = d.dept_id
JOIN Employee_Project ep
    ON e.emp_id = ep.emp_id
JOIN Project p
    ON ep.project_id = p.project_id;
```

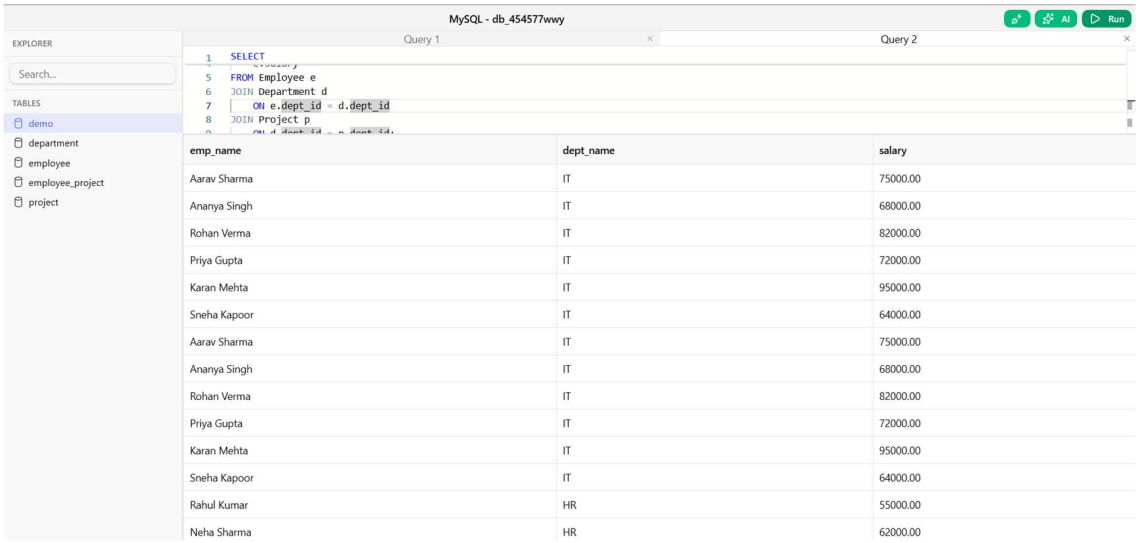
OUTPUT:



emp_name	dept_name	project_name
Aarav Sharma	IT	AI Chatbot
Ananya Singh	IT	AI Chatbot
Karan Mehta	IT	AI Chatbot
Rohan Verma	IT	Cloud Migration
Priya Gupta	IT	Cloud Migration
Sneha Kapoor	IT	Cloud Migration
Rahul Kumar	HR	Employee Training
Neha Sharma	HR	Employee Training
Amit Jain	HR	Employee Training
Pooja Verma	HR	Employee Training
Vikas Singh	HR	Employee Training
Riya Agarwal	HR	Employee Training
Arjun Patel	Finance	Financial Analysis

```
SELECT
    e.emp_name,
    d.dept_name,
    e.salary
FROM Employee e
JOIN Department d
    ON e.dept_id = d.dept_id
JOIN Project p
    ON d.dept_id = p.dept_id;
```

OUTPUT:

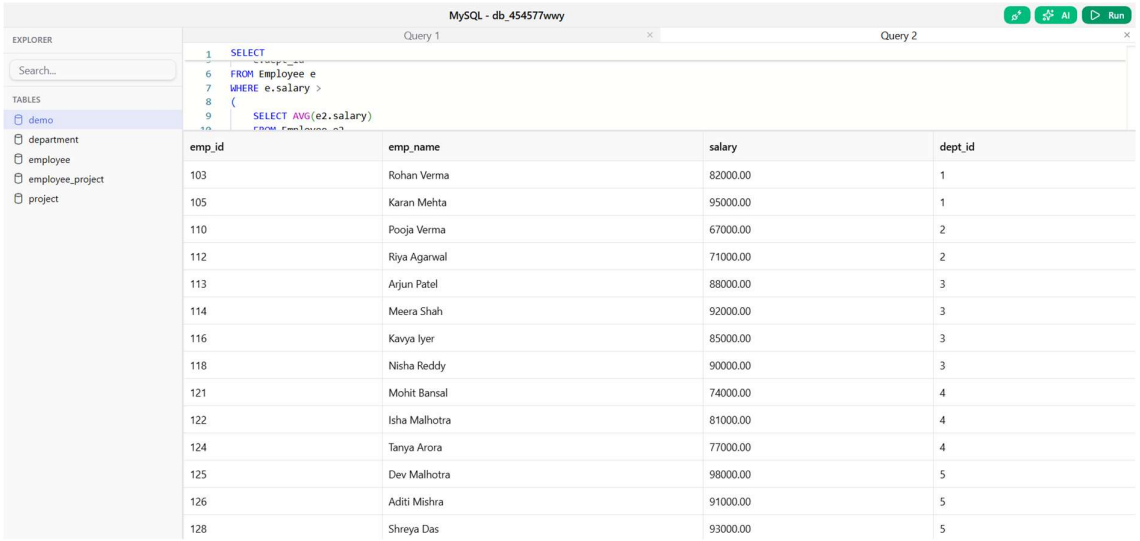


emp_name	dept_name	salary
Aarav Sharma	IT	75000.00
Ananya Singh	IT	68000.00
Rohan Verma	IT	82000.00
Priya Gupta	IT	72000.00
Karan Mehta	IT	95000.00
Sneha Kapoor	IT	64000.00
Aarav Sharma	IT	75000.00
Ananya Singh	IT	68000.00
Rohan Verma	IT	82000.00
Priya Gupta	IT	72000.00
Karan Mehta	IT	95000.00
Sneha Kapoor	IT	64000.00
Rahul Kumar	HR	55000.00
Neha Sharma	HR	62000.00

CORRELATED SUBQUERY

```
SELECT
    e.emp_id,
    e.emp_name,
    e.salary,
    e.dept_id
FROM Employee e
WHERE e.salary >
(
    SELECT AVG(e2.salary)
    FROM Employee e2
    WHERE e2.dept_id = e.dept_id
);
```

OUTPUT:

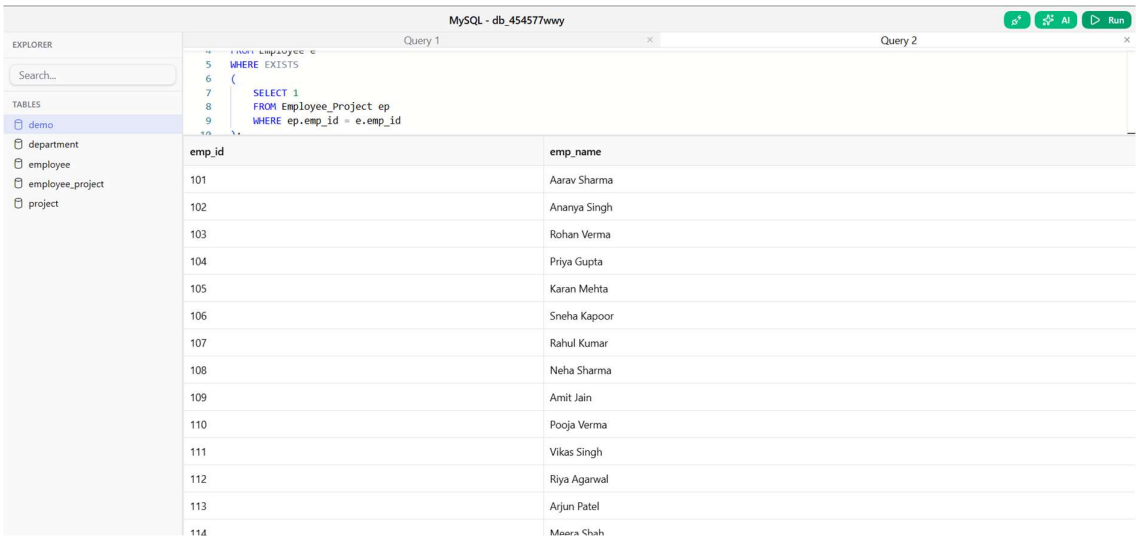


emp_id	emp_name	salary	dept_id
103	Rohan Verma	82000.00	1
105	Karan Mehta	95000.00	1
110	Pooja Verma	67000.00	2
112	Riya Agarwal	71000.00	2
113	Arjun Patel	88000.00	3
114	Meera Shah	92000.00	3
116	Kavya Iyer	85000.00	3
118	Nisha Reddy	90000.00	3
121	Mohit Bansal	74000.00	4
122	Isha Malhotra	81000.00	4
124	Tanya Arora	77000.00	4
125	Dev Malhotra	98000.00	5
126	Aditi Mishra	91000.00	5
128	Shreya Das	93000.00	5

EXISTS

```
SELECT
    e.emp_id,
    e.emp_name
FROM Employee e
WHERE EXISTS
(
    SELECT 1
    FROM Employee_Project ep
    WHERE ep.emp_id = e.emp_id
);
```

OUTPUT:



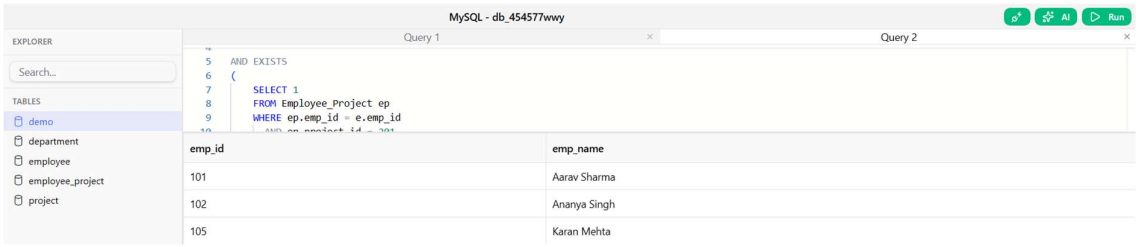
emp_id	emp_name
101	Aarav Sharma
102	Ananya Singh
103	Rohan Verma
104	Priya Gupta
105	Karan Mehta
106	Sneha Kapoor
107	Rahul Kumar
108	Neha Sharma
109	Amit Jain
110	Pooja Verma
111	Vikas Singh
112	Riya Agarwal
113	Arjun Patel
114	Meera Shah

SIMULATED INTERSECT

```
SELECT e.emp_id, e.emp_name
FROM Employee e
WHERE e.dept_id = 1

AND EXISTS
(
    SELECT 1
    FROM Employee_Project ep
    WHERE ep.emp_id = e.emp_id
      AND ep.project_id = 201
);
```

OUTPUT:

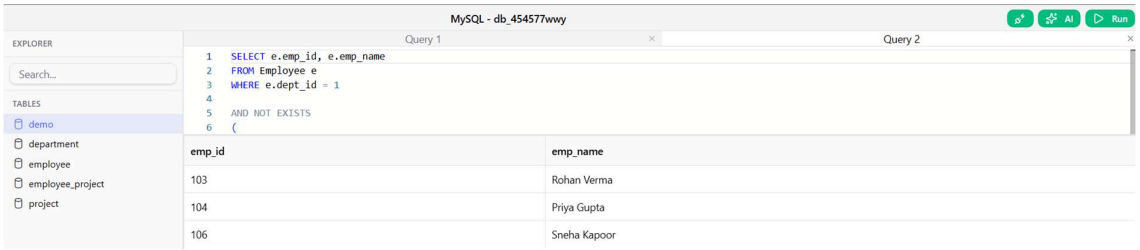


SIMULATED EXCEPT

```
SELECT e.emp_id, e.emp_name
FROM Employee e
WHERE e.dept_id = 1

AND NOT EXISTS
(
    SELECT 1
    FROM Employee_Project ep
    WHERE ep.emp_id = e.emp_id
      AND ep.project_id = 201
);
```

OUTPUT:

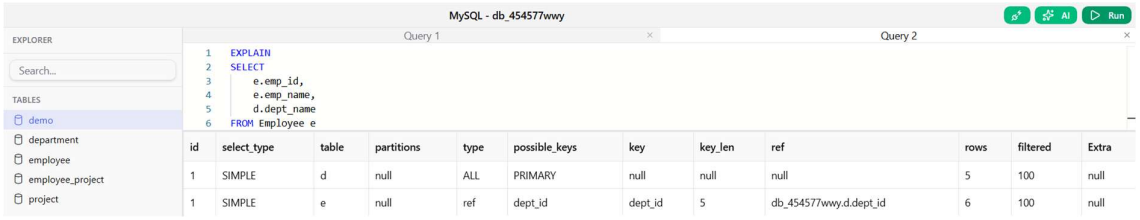


COMPARING EXECUTION PLANS USING EXPLAIN

EXPLAIN – INNER JOIN

```
EXPLAIN
SELECT
    e.emp_id,
    e.emp_name,
    d.dept_name
FROM Employee e
INNER JOIN Department d
    ON e.dept_id = d.dept_id;
```

OUTPUT:

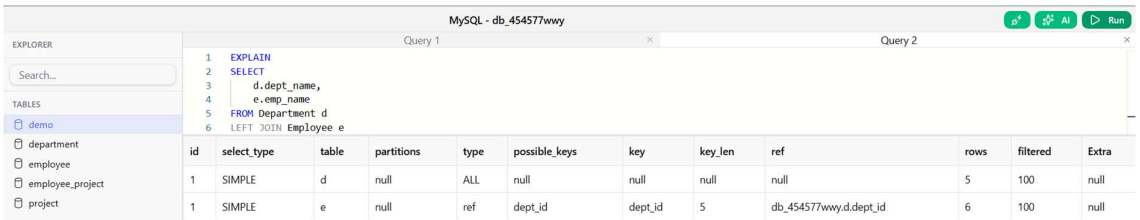


id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	d	null	ALL	PRIMARY	null	null	null	5	100	null
1	SIMPLE	e	null	ref	dept_id	dept_id	5	db_454577wwy.d.dept_id	6	100	null

EXPLAIN – LEFT JOIN

```
EXPLAIN
SELECT
    d.dept_name,
    e.emp_name
FROM Department d
LEFT JOIN Employee e
    ON d.dept_id = e.dept_id;
```

OUTPUT:

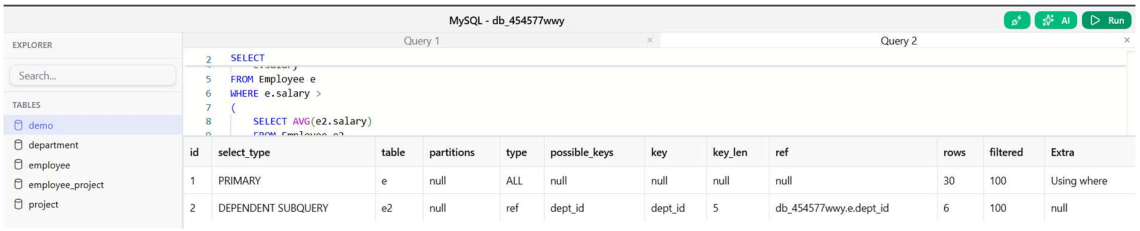


id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	d	null	ALL	null	null	null	null	5	100	null
1	SIMPLE	e	null	ref	dept_id	dept_id	5	db_454577wwy.d.dept_id	6	100	null

EXPLAIN – CORRELATED SUBQUERY

```
EXPLAIN
SELECT
    e.emp_name,
    e.salary
FROM Employee e
WHERE e.salary >
(
    SELECT AVG(e2.salary)
    FROM Employee e2
    WHERE e2.dept_id = e.dept_id
);
```

OUTPUT:

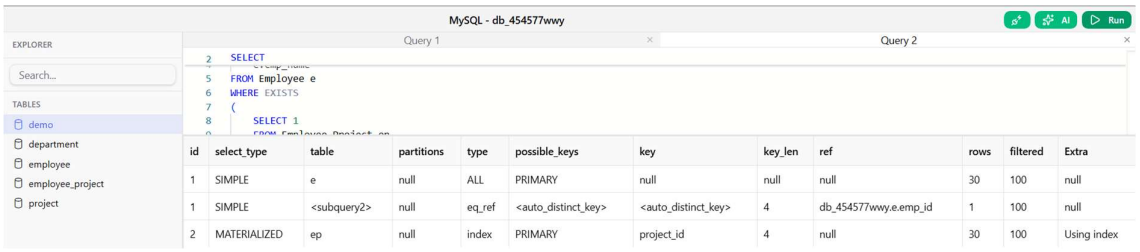


id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	PRIMARY	e	null	ALL	null	null	null	null	30	100	Using where
2	DEPENDENT SUBQUERY	e2	null	ref	dept_id	dept_id	5	db_454577wwy.e.dept_id	6	100	null

EXPLAIN – EXISTS

```
EXPLAIN
SELECT
    e.emp_id,
    e.emp_name
FROM Employee e
WHERE EXISTS
(
    SELECT 1
    FROM Employee_Project ep
    WHERE ep.emp_id = e.emp_id
);
```

OUTPUT:



id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	e	null	ALL	PRIMARY	null	null	null	30	100	null
1	SIMPLE	<subquery2>	null	eq_ref	<auto_distinct_key>	<auto_distinct_key>	4	db_454577wwy.e.emp_id	1	100	null
2	MATERIALIZED	ep	null	index	PRIMARY	project_id	4	null	30	100	Using index